.5 the NO x emission performance of the engine combined with the SCR is verified by a confirmation test in accordance with the procedure in paragraph 7.5 of these guidelines.

6.2 Verification test procedures for an engine

6.2.1 The purpose of the test of an engine is to establish the emission values for use in section 6.4 of these guidelines. These measurements should be in accordance with chapter 5 of the NTC 2008.

6.2.2 Paragraph 5.9.8.1 of the NTC 2008 requires engine conditions to be measured at each mode point, for an engine system. This equally applies in the case of an engine fitted with SCR. Additionally, exhaust gas temperature at the intended inlet of the SCR chamber should be determined and recorded in the test report as required by section 5.10 of the NTC 2008.

6.3 Test procedures for SCR chambers

6.3.1 General

6.3.1.1 The SCR chamber for validation testing may be either a full scale SCR chamber or a

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SOURCE: MEPC.198(62).pdf
NTC 2008.

6.3 Test procedures for SCR chambers

6.3.1 General

6.3.1.1 The SCR chamber for validation testing may be either a full scale SCR chamber or a scaled version. A SCR chamber should demonstrate the reduction in NO x concentrations (ppm) expected in exhaust gas measured in section 6.2 of these guidelines. Therefore, NO x reduction rate of the SCR chamber should be determined for each individual mode point. Where undertaken on a scaled version of the SCR chamber the scaling process should be validated to the satisfaction of the Administration.

6.3.2 Test conditions at each mode point

6.3.2.1 Exhaust gas, catalyst, reductant and an injection system should satisfy the following conditions at each mode point:

.1 Exhaust gas flow
Exhaust gas flow rate for the test should be scaled accordingly to account for the dimension of the catalyst model.

.2 Exhaust gas component
Exhaust gas for the test should either be diesel engine exhaust gas or simulated gas.
RESOLUTION MEPC.198(62)
Adopted on 15 July 2011
2011 GUIDELINES ADDRESSING ADDITIONAL ASPECTS TO THE NOx TECHNICAL

CODE 2008 WITH REGARD TO PARTICULAR REQUIREMENTS RELATED TO MARINE

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SOURCE: MEPC.198(62).pdf
simulated gas.
RESOLUTION MEPC.198(62)
Adopted on 15 July 2011
2011 GUIDELINES ADDRESSING ADDITIONAL ASPECTS TO THE NOx TECHNICAL
CODE 2008 WITH REGARD TO PARTICULAR REQUIREMENTS RELATED TO MARINE
DIESEL ENGINES FITTED WITH SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 62/24
Annex 6, page 8

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Where diesel exhaust gas is used it should correspond, in terms of concentrations, to the exhaust gas in section 6.2 of these guidelines, in terms of NO x, O2, CO2, H2O, and SO 2 (±5% of the required concentration for each emission species).

Where simulated gas is used it should correspond, in terms of concentrations, to the exhaust gas in section 6.2 of these guidelines, in terms of NO, NO 2, O 2, CO 2, H 2O, and SO 2 (±5% of the required concentration for each emission species) balance N2.

.3 Exhaust gas temperature
The temperature of exhaust gas used for the test should correspond to the temperatures obtained from testing in section 6.2 of these guidelines, ensuring that the SCR chamber is activated at every load point, other than as provided for by 3.1.4 of the NTC 2008, and that no ammonia bisulphate

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SOURCE: MEPC.198(62).pdf
temperatures obtained from testing in section 6.2 of these guidelines, ensuring that the SCR chamber is activated at every load point, other than as provided for by 3.1.4 of the NTC 2008, and that no ammonia bisulphate formation, or reductant destruction, takes place.

.4 Catalyst blocks and AV,SV value
The catalyst blocks used in the test should be representative of the catalyst blocks to be used in the SCR chamber in service. AV,SV or LV value should, in the case of full scale tests, be within a range of ±20% of the required value as obtained in testing from section 6.2 of these guidelines. In the case of scaled tests it should correspond to the above.

.5 Reductant
The reductant concentration should be representative of the reductant concentration in the exhaust gas during actual operation.

6.3.3 Stability for measurement

6.3.3.1 All measurements should be re corded after they have stabilized.

6.3.4 List of data to be derived from the model

6.3.4.1 Operating data which is to be given in the Technical File should be derived from the modelling process or otherwise justified.

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SOURCE: MEPC.198(62).pdf
6.3.4 List of data to be derived from the model

6.3.4.1 Operating data which is to be given in the Technical File should be derived from the modelling process or otherwise justified.

6.3.4.2 Exhaust gas analysers should be in accordance with appendix III and appendix IV of the NTC 2008 or otherwise to the satisfaction of the Administration.

6.3.5 Test report for SCR chamber

6.3.5.1 Data recorded under paragraph 6.3.1.1 of these guidelines should be recorded in the test report as required by section 5.10 of the NTC 2008.

6.4 Calculation of the specific emission

6.4.1 The NO x emission value of the engine system fitted with SCR should be calculated as follows:

n i
i F
F
n i
i mgasi
x
i
ii
W P
W q
1 i
1 100100gas

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Where: i = NO x reduction rate (%) derived in accordance with section 6.3 of these guidelines;

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SOURCE: MEPC.198(62).pdf
DIESEL ENGINES FITTED WITH SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 62/24
Annex 6, page 9

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Where: i = NO x reduction rate (%) derived in accordance with section 6.3 of these guidelines;

imgasq = Mass flow of NO x gas measured in accordance with section 6.2 of these guidelines;

iFW = Weighting factor;
iP = Measured power at individual mode points in accordance with section 6.2 of these guidelines.

The weighting factors and number of modes (n) used in above calculation shall be according to the provisions of section 3.2 of the NTC 2008.